

and the r.m.s.e. is defined as

$$\text{r.m.s.e.} = \sqrt{(f - o)^2}, \quad (8)$$

where σ_f and σ_o are the standard deviations of the observations and forecast, respectively.

The Fisher z transformation was used to test statistical significance of the ACC differences as follows:

$$Z = 0.5 \frac{\ln\left(\frac{1+r_1}{1-r_1}\right) - \ln\left(\frac{1+r_2}{1-r_2}\right)}{\sqrt{\frac{1}{n_1-3} + \frac{1}{n_2-3}}}, \quad (9)$$

where r_1 and r_2 are the correlation coefficients, n_1 and n_2 are the sample sizes of the first and second group samples. The absolute value $|Z|$ was then compared against a critical value from the t distribution for a two-tailed test. We rejected the null hypothesis that the two correlations were not significantly different at 90% confidence level if $|Z|$ exceeds the critical value.

The XRO reforecasting experiments based on CMIP model output

We analysed monthly mean SST and three-dimensional ocean temperature fields from 91 CMIP5 and CMIP6 historical simulations (Supplementary Table 5). All model outputs were re-gridded to a common $1^\circ \times 1^\circ$ horizontal resolution using bilinear interpolation. The monthly anomaly fields were calculated by removing the monthly climatology for the period of 1900–1999 and the quadratically detrended over the full 100 year period.

Using the linear and nonlinear operators trained solely on CMIP model m output for 1900–1999, we conducted retrospective 21 month forecasts with initial conditions from the observations for the period of January 1979–October 2023 (referred to as XRO^m). To understand the impacts of model biases on ENSO dynamics and its coupling with other modes, we also conducted sensitivity experiments by correcting the different components of the linear and nonlinear operators with the observed parameters (Extended Data Table 2). For example, the experiment XRO_L^m is the same as XRO^m, but with the linear operator $\mathbf{L}$ being replaced by the observed $\mathbf{L}$ operator. The difference XRO_L^m – XRO^m was used to isolate the effect of correcting model m 's linear dynamics biases. Similarly, the experiments XRO_{ENSO}^m, XRO_{C₁}^m and XRO_{C₂}^m were conducted to isolate the impacts of model m 's biases on the internal linear ENSO dynamics, the coupling feedback to ENSO parameters and ENSO teleconnection dynamics, respectively.

The XRO2 ENSO types and pantropical SSTA forecasts

The extra XRO model (referred to XRO2) was set up to predict different types of El Niño (that is, ENSO diversity). We introduced two SSTA indices in the state vectors of ENSO, that is, the Niño3 index (SSTAs averaged over $150^\circ\text{--}90^\circ\text{W}$, $5^\circ\text{S--}5^\circ\text{N}$) and Niño4 index (SSTAs averaged over $160^\circ\text{E--}150^\circ\text{W}$, $5^\circ\text{S--}5^\circ\text{N}$): $\mathbf{X}_{\text{ENSO}} = [T_{\text{Niño3}}, T_{\text{Niño4}}, h]$ instead of using Niño3.4. The quadratic nonlinearities $b_1 T_{\text{Niño3}}^2 + b_2 T_{\text{Niño3}} h$ were only incorporated into the SSTA equation of $T_{\text{Niño3}}$, in the presence of a strong asymmetry of Niño3 index, whereas there was a less-pronounced asymmetry of Niño4 index: $\mathbf{N}_{\text{ENSO}} = [b_1 T_{\text{Niño3}}^2 + b_2 T_{\text{Niño3}} h, 0, 0]$. All other terms were the same as the standard XRO model. Using the operator parameters estimated from the ORAS5 reanalysis for 1979–2022, we conducted similar retrospective 21 month forecasts for the period of January 1979–October 2023. The hindcast skills of Niño3 and Niño4 indices are better than those from the NMME dynamical models and comparable to the AI model. The forecasts of Niño3 and Niño4 indices were used to define the El Niño types in terms of the EP-type, CP-type and mixed-type, following refs. 8,100. The unified complex ENSO index (UCEI) is defined as

$$\text{UCEI} = (N_3 + N_4) + (N_3 - N_4)i = re^{i\theta}, \quad (10)$$

where

$$r = \sqrt{(N_3 + N_4)^2 + (N_3 - N_4)^2}, \quad (11)$$

and

$$\theta = \begin{cases} \arctan \frac{N_3 - N_4}{N_3 + N_4} & \text{when } N_3 + N_4 > 0 \\ \arctan \frac{N_3 - N_4}{N_3 + N_4} - \pi & \text{when } N_3 + N_4 < 0 \end{cases} \quad (12)$$

where N_3 and N_4 denote the Niño3 and Niño4 indices, respectively. The El Niño type is determined from θ as follows:

$$\begin{cases} 15^\circ \leq \theta < 90^\circ & \text{EP El Niño} \\ -15^\circ \leq \theta < 15^\circ & \text{Mixed El Niño} \\ -90^\circ \leq \theta < -15^\circ & \text{CP El Niño} \end{cases} \quad (13)$$

We also conducted out-of-sample XRO2 ENSO type reforecasts by fitting on 1950–1990 with the multi-products ensemble mean indices and verifying on 1991–2022 (Supplementary Table 3).

With the forecasted ten SSTA indices, the pantropical SSTA ($30^\circ\text{S--}30^\circ\text{N}$) at each grid point (SSTA_{*j*}) can be predicted using the seasonal regression model:

$$\text{SSTA}_j = c_0 X + A_c X \cos \omega t + A_s X \sin \omega t + B_c X \cos 2\omega t + B_s X \sin 2\omega t, \quad (14)$$

where c_0 , A_c , A_s , B_c and B_s have ten coefficients associated with each SSTA index, respectively. We also conducted the cross-validated XRO2 forecasts and pantropical SSTA forecast by excluding 3 year data and trained XRO2 operators and SSTA regression coefficients, then forecasts for each month during the years not included in the model fitting.

Further details are provided in the Supplementary Information, relying on refs. 101–113.

Data availability

Datasets used in this paper are freely available. Observational data have links in Supplementary Table 2; NMME: <https://iridl.ldeo.columbia.edu/SOURCES/.Models/.NMME/>; 3D-Geoformer ENSO AI model forecast: <http://msdc.qdio.ac.cn/data/metadata-special-detail?id=1602252663859298305>; CESM1 LENS: <https://www.cesm.ucar.edu/community-projects/lens/data-sets>; CESM2 LENS: <https://www.cesm.ucar.edu/community-projects/lens2/data-sets>; MPI-ESM LENS: <https://esgf-data.dkrz.de/projects/mpi-ge/>; CMIP5 outputs: <https://esgf-node.llnl.gov/projects/cmip5/>; and MIROC6 LENS and CMIP6 outputs: <https://esgf-node.llnl.gov/projects/cmip6/>. All the map figures (Fig. 1a,c,d and Supplementary Figs. 1, 2 and 14) were generated using Python Cartopy v.0.22.0 (<https://doi.org/10.5281/zenodo.1182735>) (ref. 114). The source data for figures in the main text are available at <https://doi.org/10.5281/zenodo.10951443> (ref. 115).

Code availability

The XRO model code is deposited at <https://doi.org/10.5281/zenodo.10681114> (ref. 116). The code to calculate the predictive skill is available at <https://github.com/pangeo-data/climpred>.

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